Nihar Chauhan
UIN: 434007224

Date: 04/28/2025

CSCE 685 - Directed Studies

AI News Hub - Project Report

Abstract

The AI News Hub is an intelligent platform designed to aggregate, summarize, and translate global news content in real time using advanced artificial intelligence. It integrates the NewsAPI for real time news retrieval and OpenAI's GPT-4 for generating concise, insightful summaries. To ensure accessibility across linguistic boundaries, it supports multi-language translation. The system is optimized with asynchronous processing and a custom caching mechanism, resulting in significant performance gains. A modern, user-friendly interface built using Gradio enables seamless interaction for users to explore news across various categories. The project gives a scalable, accessible solution for staying informed in today's world. The platform helps users avoid feeling overwhelmed by too much information and makes news easier to understand for everyone, no matter what language they speak. This makes it not just a tool for casual users but a potential asset for educators, journalists, and researchers looking for rapid insights from global media.

Introduction

The global media landscape is filled with information, making it very difficult for users to keep up with significant events, especially across different languages and time zones. Most traditional news apps either just show a bunch of headlines with no context or provide you with long articles. They often miss that personal touch and don't make it easy to quickly get the main idea. The AI News Hub bridges this gap by using AI technologies to deliver brief yet meaningful insights on news across different domains and in various languages.

This platform was built to overcome language and content length barriers in news consumption, provide real time, AI curated news summaries, and offer a visually appealing and intuitive interface to enhance user engagement.

Main Idea

The main idea behind AI News Hub lies in making the experience of reading and understanding news faster, smarter, and more accessible for everyone. In a world where news is constantly flowing and people have limited time, the goal is simple to help users stay informed without the need to scroll through long articles.

Instead of overwhelming users with full length stories, the platform uses GPT-4 to automatically generate short, easy to digest summaries that highlight the most important facts, events, and people involved. This helps users quickly grasp what is happening in the world.

What truly sets AI News Hub apart is its ability to translate content into different languages, making news accessible to people from different backgrounds and regions. Whether someone speaks English, Spanish, or Hindi, they can get the same quality summaries in their preferred language, bridging the language gap in global news.

To bring this idea to life, three major goals were focused while developing it:

1. End to End Automation

The entire process which is from fetching the latest headlines, to summarizing them using AI, to translating the output is fully automated. This means that users don't have to wait or manually search. Everything happens behind the scenes in just a few seconds.

2. Real Time Performance

The backend was built with asynchronous programming to handle multiple articles at once. With the help of a smart caching system, this ensures that the platform remains fast and efficient, even as the number of users or news stories grows.

3. Simple and Clean User Interface

Using Gradio, the frontend was designed to be friendly and intuitive. Users can easily pick a news category, choose a language, and instantly see a list of summarized articles. All of them are styled to be visually appealing and easy to read.

Method

To bring the AI News Hub idea to life, the platform was developed step by step, integrating intelligent automation with user-friendly design. This section combines the methodology with the technical implementation, showing how each part works together to create a fast, responsive, and inclusive news experience.

1. Data Acquisition using Real-Time News Retrieval

The first step in the system is gathering the latest news articles from around the world. This is achieved using the NewsAPI which is a reliable source that aggregates news from a wide range of reputable publishers.

There are different categories supported in this application like Business, Entertainment, Sports, General, Health, Science, and Technology.

Each article includes:

- **Title**

- **Short description**
- **Source name**
- **Direct URL to the full article**
- **Image preview (if available)**

Key Function: fetch_global_headlines()
This function connects to NewsAPI and retrieves articles based on the selected category. It parses the results and returns a clean, structured list of article objects.

2. Summarization and Translation using GPT-4 Integration

Once the articles are fetched, the next step is to simplify the content and make it multilingual.

First step is summarization. OpenAI's GPT-4 is used to process the main points of each article into a few clear sentences, capturing the key events, people involved, and why the story matters.

Second step is translation. If the user chooses a language other than English, GPT-4 is used again to translate both the article's title and its summary into the selected language while maintaining tone and meaning.

Key Functions:
summarize_article() - Sends a well defined prompt to GPT-4 and returns a short, readable summary.

translate_text() - Translates the summary and title into the user's chosen language.

Technology Used:
OpenAI GPT-4 is used with the AsyncOpenAI client for asynchronous operations, which is crucial for reducing latency when processing many articles concurrently.

3. Asynchronous Backend for Speed and Responsiveness

To keep the platform fast and responsive, especially when dealing with dozens of articles, the backend uses asynchronous programming.

Technology Used:
Python is used with asyncio and aiohttp for non-blocking API calls and background processing.

Summarization, translation, and article fetching run concurrently to ensure efficient parallel processing and avoiding bottlenecks.

Key Function: news_aggregator_async()
It acts as the main functionality as it initializes the OpenAI client, fetches news articles, launches asynchronous processing for each article using process_article(), and collects and formats results as styled HTML for display.

4. **Caching Mechanism for efficiency and cost saving**

Since GPT-4 operations can be resource consuming, a custom caching system was developed to avoid redundant work.

Cache type which was used is the Least Recently Used (LRU) cache implemented using Python's OrderedDict.

The key functionality of this is that if a summary or translation has already been created for an article in a particular language, it is pulled from the cache rather than regenerated.

Technology Used: Custom SimpleCache class
This significantly cuts down API calls, improves response times, and lowers operational costs.

5. **Frontend Integration for user interface with Gradio**

All the backend intelligence is wrapped in a modern, clean user interface created using Gradio.

It provides key user features such as:

- **Dropdown menus for selecting news category and preferred language**
- **A button to fetch summarized articles**

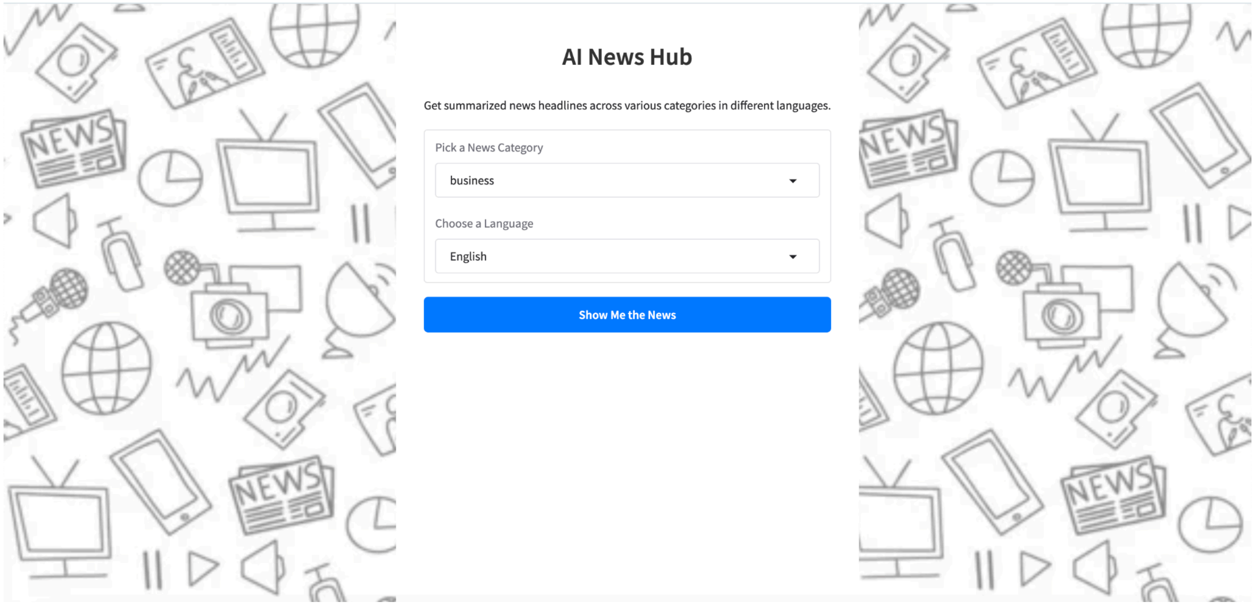
Display of summarized content with:
- **Clickable headlines**
- **News source name**
- **Summary text**
- **Preview image (if available)**

Technology Used:
Gradio for frontend and UI elements and also for seamless linking of Python backend functions with UI controls.

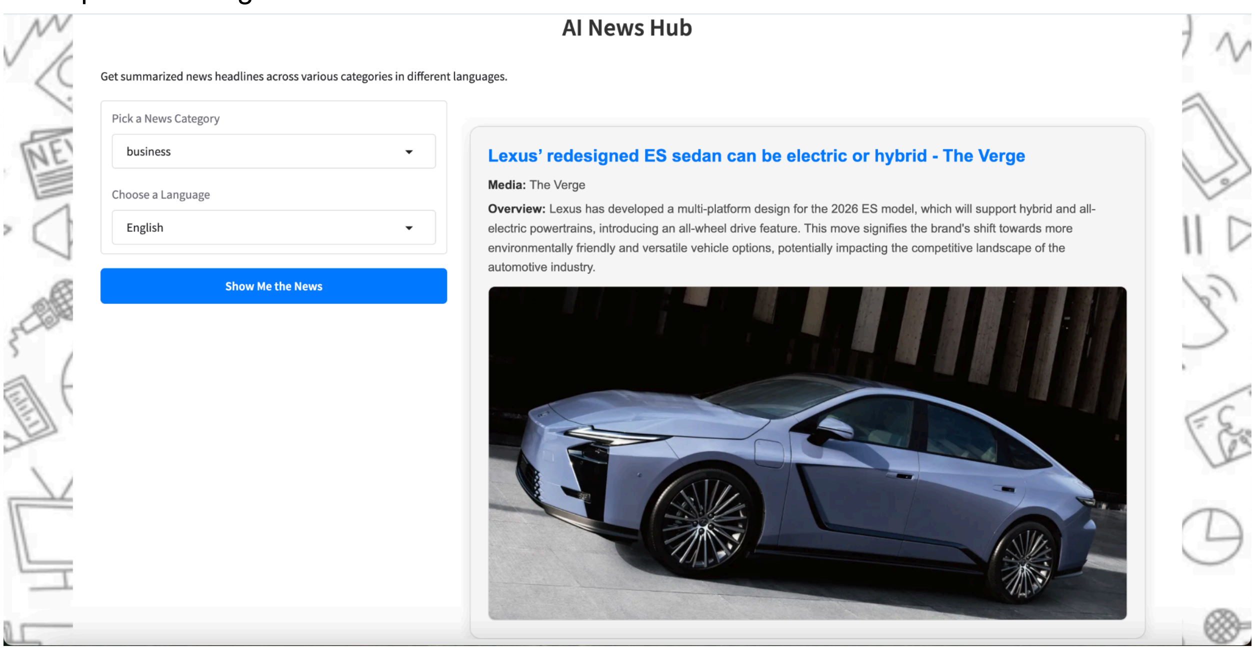
Results

The AI News Hub project delivered excellent performance in terms of speed, accuracy, and user experience.

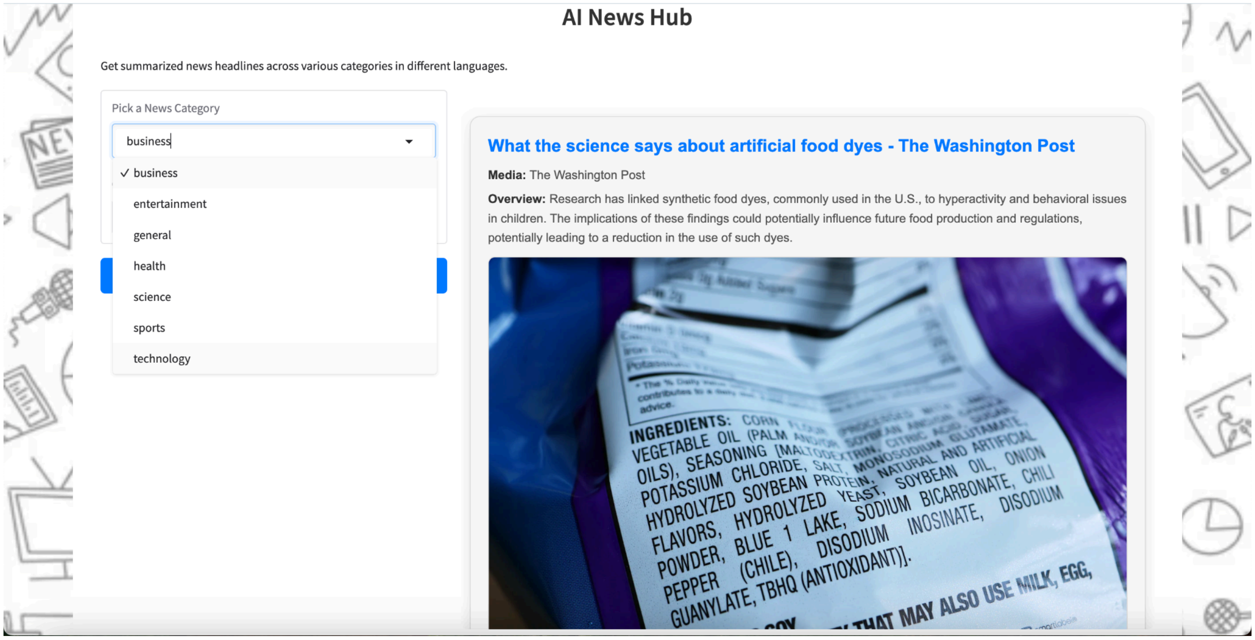
This is the main page of the AI News Hub application. It offers a dropdown where you can pick a news category and another dropdown lets you choose a language. The “Show Me the News” button would list out the news articles based on the news category and language a user has selected.



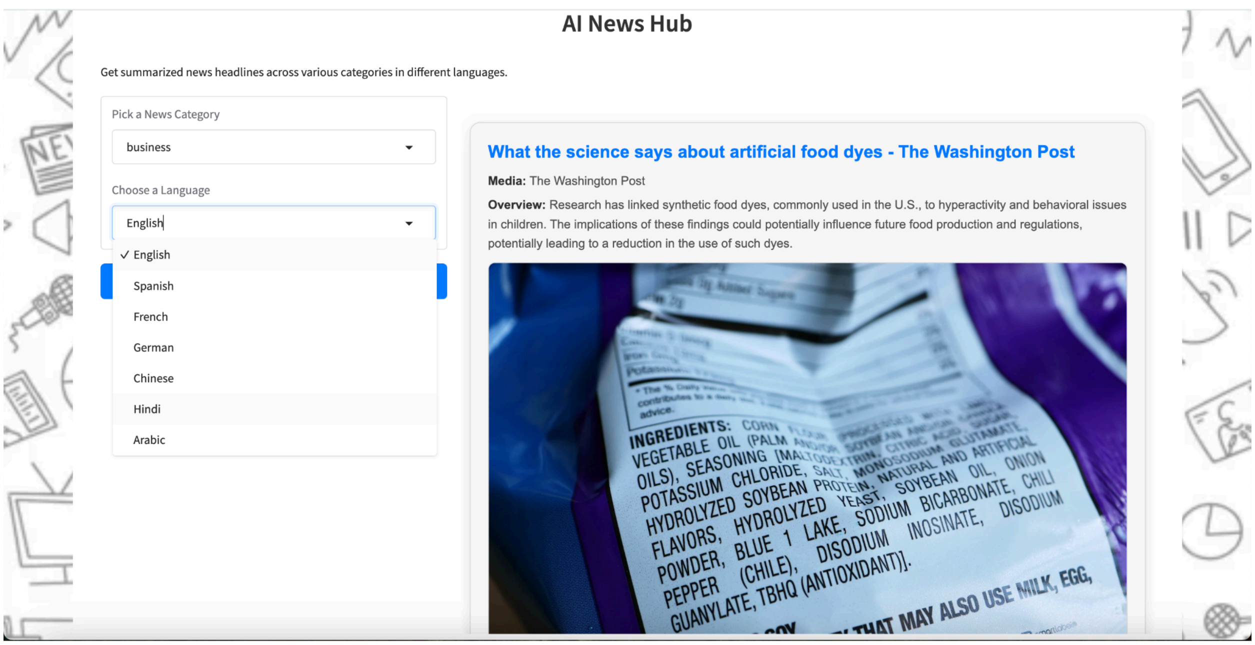
You get 15 scrollable concise news articles for business news categories and in English language. The news article consists of clickable headlines, news source name, summary text, and a preview image.



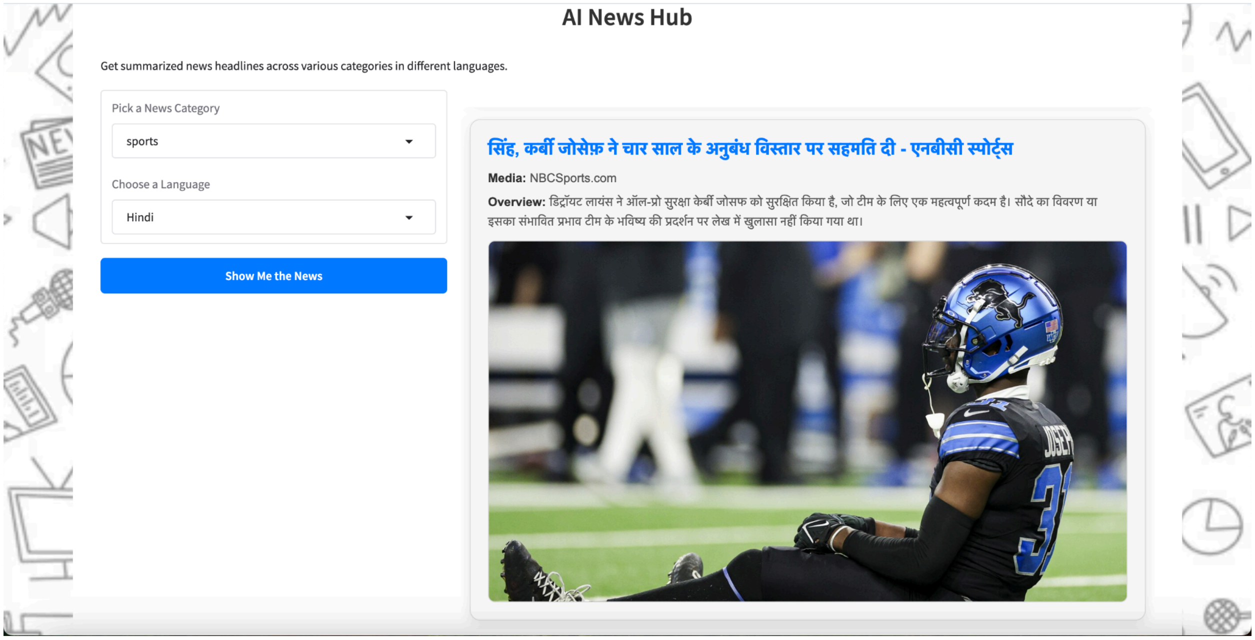
Here are the different categories supported in this application in the dropdown like Business, Entertainment, General, Health, Science, Sports and Technology.



These are the different languages supported in this application in the dropdown like English, Spanish, French, German, Chinese, Hindi, Arabic.



This is an example news article where the news category is “sports” and language is “Hindi”. The news article card shows entirely in Hindi language.



1. Fast and Reliable Summaries

One of the most critical performance aspects was how quickly the platform could generate summaries. This is because of the use of asynchronous programming and a caching system, the average time to summarize each article was consistently under 2 seconds when summaries were not cached and nearly instant when retrieved from the cache. This made the user experience smooth and efficient, especially when processing multiple articles at once.

2. High-Quality and Accurate Translations

The translation feature using GPT-4, showed impressive contextual understanding. When summaries and headlines were translated into other languages, the results maintained meaningful accuracy, no grammatical correctness, and natural-sounding phrasing appropriate for each language. Along with English, testing with French, Hindi, and Spanish showed promising outcomes, proving the platform is ready for full multilingual expansion.

3. Scalability and Performance Optimization

The combination of asynchronous tasks and a lightweight caching strategy allowed the platform to handle up to 30-50 articles in a matter of seconds without freezing or lagging. Even with multiple users querying different categories, the system handled the load effectively. This proves the architecture is ready for real-world scalability with minimal server resources.

4. Real-World Use Case Demonstration

To showcase its real-world utility, a sample test was conducted:

Use Case:

A user selects “**Business**” as the news category and “**English**” as the preferred language.

Within just a few seconds, the system fetched and processed 10-15 articles. Each article was displayed with a headline, media source, a short summary, and image.

Summaries were clearly written and easy to scan.

When switching to another language (e.g., Hindi or French), the summaries and titles were automatically translated which demonstrated the multilingual capability of the platform without requiring any extra effort from the user.

Technical Learnings:

Mastering asynchronous programming patterns in Python, including async/await, aiohttp, and asyncio.gather, greatly enhances the ability to handle concurrent tasks efficiently. Effective use of GPT-4 for both summarization and language translation broadens the scope of projects. Understanding performance bottlenecks and implementing caching strategies reduces API costs and latency, improving overall system performance. Exploring Gradio’s advanced layout capabilities and CSS customization enables the creation of more intuitive and visually appealing user interfaces. Using AI to summarize news helps people deal with too much information. Adding support for different languages makes the app more inclusive and useful for people around the world. Combining backend optimization with intuitive frontend design maximizes usability and performance which delivers a seamless user experience.

Conclusion:

This project shows that building a real-time, AI-powered news platform is both feasible and effective by combining NewsAPI, GPT-4, and Gradio. The system uses intelligent automation and asynchronous processing to quickly turn raw news content into concise, informative summaries, making global news easier to access.

It uses built-in translation tools to break language barriers, making it easy for people who speak different languages to use the platform. A smart caching mechanism ensures fast performance and reduces API usage while maintaining high-quality output.

The modular and scalable design allows for rapid processing of large volumes of articles, and the user-friendly frontend ensures a smooth experience. These choices prove the platform’s practicality for real-time use and its potential to become a more advanced news assistant.

The AI News Hub shows the power of combining advanced language models with modern web technologies to create a lightweight, scalable, and user-friendly solution for efficient news consumption in today’s digital world.

GitHub URL: **https://github.com/niharchauhan/AI-News-Hub**